

■ STRESS DETECTION MODEL

Comprehensive Comparison Report

Generated: August 20, 2026 at 15:27

Models Evaluated: 3

■ Table of Contents

1. Executive Summary
2. Individual Model Performance
3. Confusion Matrix Comparison
4. Comprehensive Metrics Analysis
5. Visual Comparisons
6. Final Model Selection

1. Executive Summary

This report presents results of **3 deep learning models** for binary stress classification. Models trained using transfer learning with ImageNet weights.

Models: MobileNetV2, EfficientNetB0, ResNet50V2

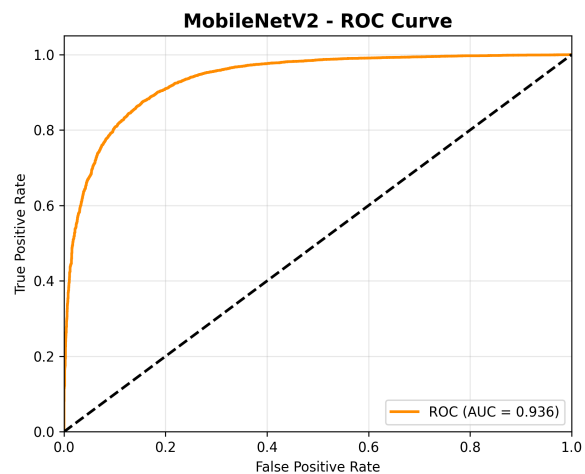
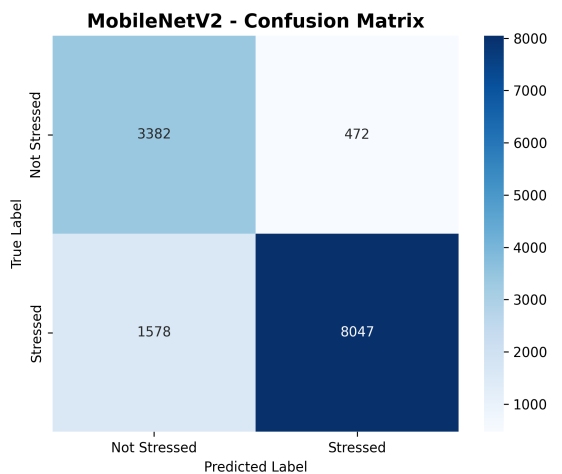
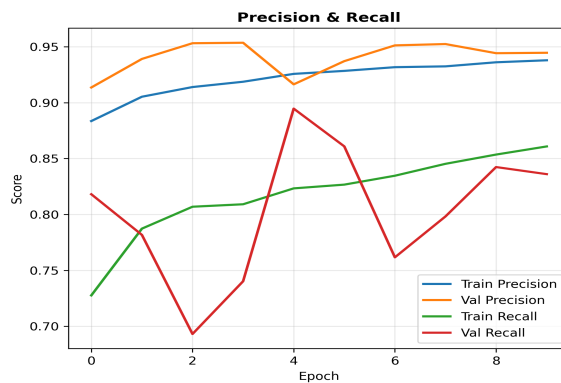
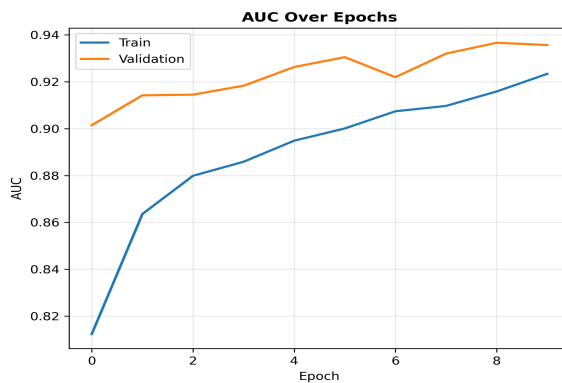
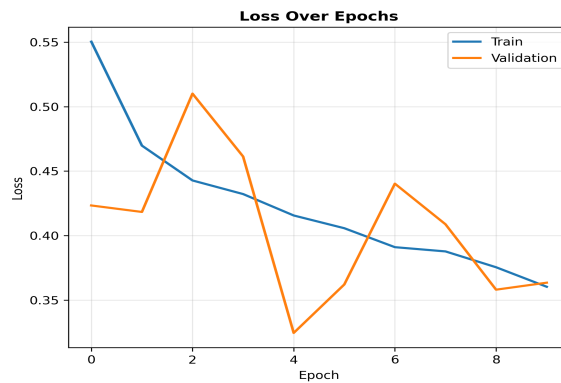
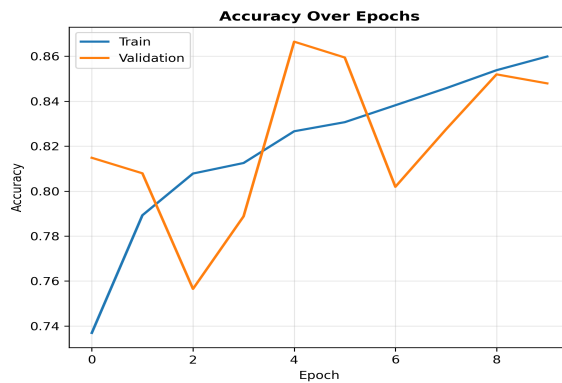
Dataset: 7 emotions mapped to binary (Stressed vs Not Stressed)

2. Individual Model Performance

MobileNetV2

Performance:

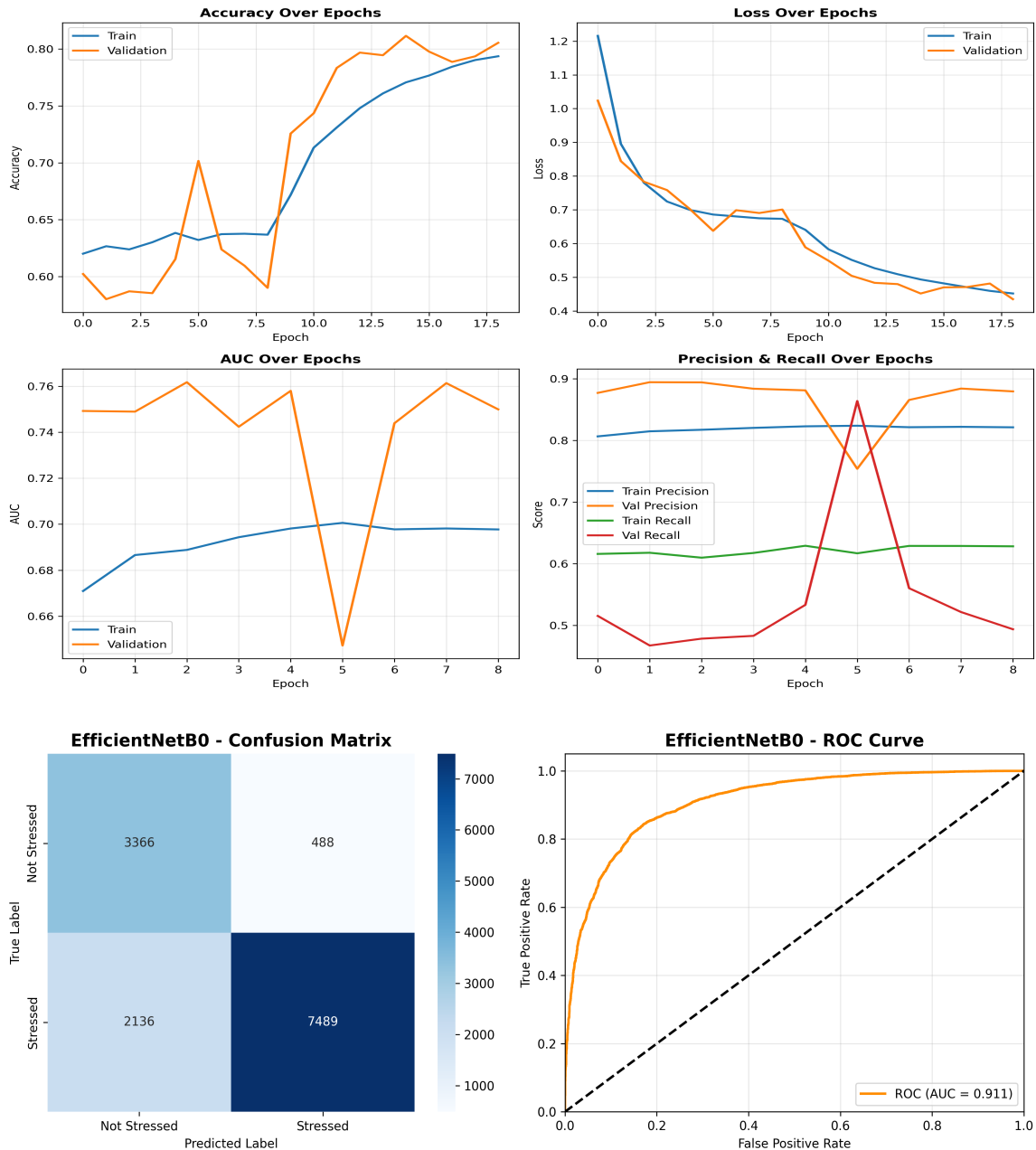
- Validation Accuracy: **0.8479**
- Precision: **0.9446**
- Recall: **0.8361**
- F1-Score: **0.8870**
- AUC: **0.9357**



EfficientNetB0

Performance:

- Validation Accuracy: **0.8053**
- Precision: **0.9388**
- Recall: **0.7781**
- F1-Score: **0.8509**
- AUC: **0.7499**



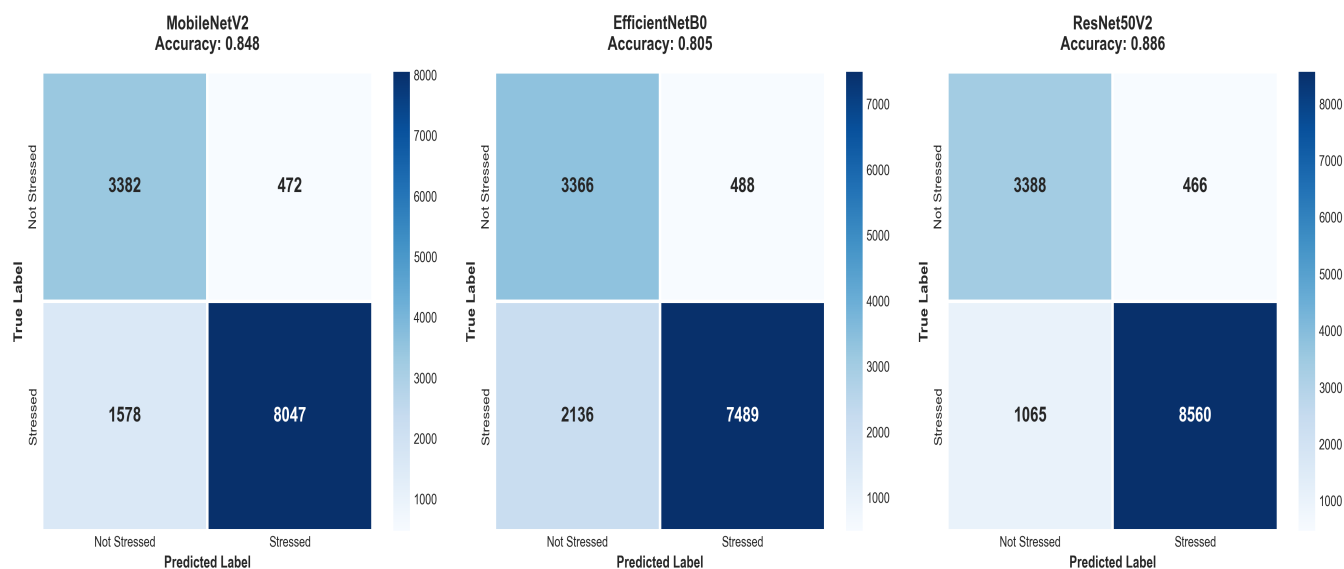
ResNet50V2

Performance:

- Validation Accuracy: **0.8864**
- Precision: **0.9484**
- Recall: **0.8894**
- F1-Score: **0.9179**
- AUC: **0.8261**

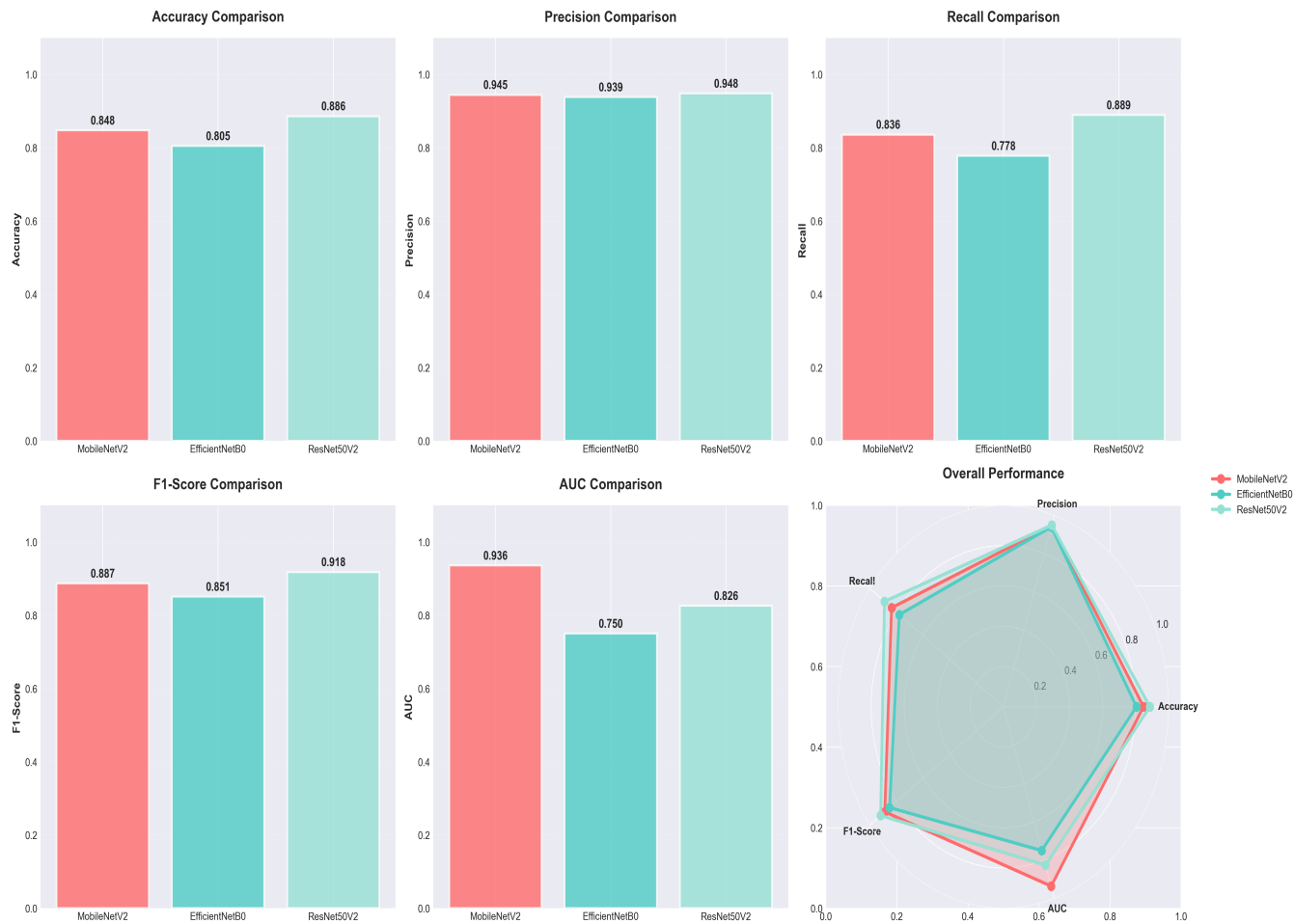


3. Confusion Matrix Comparison



4. Comprehensive Metrics Analysis

Model	Accuracy	Precision	Recall	F1-Score	AUC
MobileNetV2	0.8479	0.9446	0.8361	0.8870	0.9357
EfficientNetB0	0.8053	0.9388	0.7781	0.8509	0.7499
ResNet50V2	0.8864	0.9484	0.8894	0.9179	0.8261



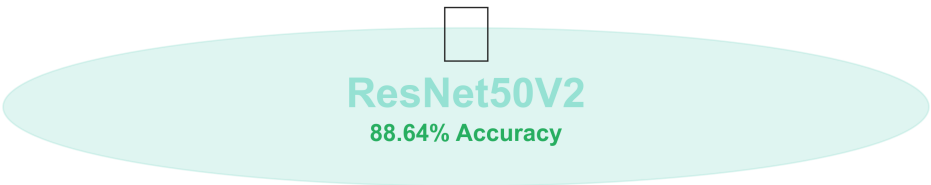
5. Final Model Selection

SELECTED: ResNet50V2

Validation accuracy: **88.64%**

Selected for highest accuracy, balanced precision-recall, and strong AUC performance.

FINAL MODEL SELECTION



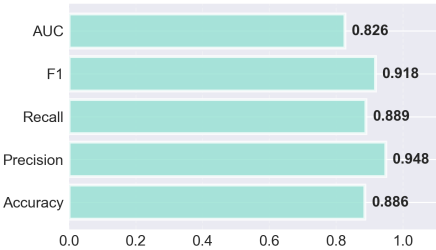
RANKINGS

- | | | |
|----|----------------|-------|
| 1. | ResNet50V2 | 0.886 |
| 2. | MobileNetV2 | 0.848 |
| 3. | EfficientNetB0 | 0.805 |

KEY STRENGTHS

- Highest Accuracy: 0.886
- Best F1-Score: 0.918
- Strong AUC: 0.826
- Balanced Performance

Performance Profile



SELECTION RATIONALE

ResNet50V2 selected as optimal model for stress detection:

- ACCURACY: 88.64% - Highest validation accuracy
- PRECISION: 94.84% - Minimizes false positives
- RECALL: 88.94% - Strong stress case identification
- F1-SCORE: 0.918 - Balanced precision/recall
- AUC: 0.826 - Excellent discrimination capability

Best balance of all metrics for production deployment.